GLOW

Habit Tracker

Mobile Habit Tracker

University of Msila · Mobile Applications Course

Flutter · SQL · Provider · MVVM Architecture

2025 / 2026

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01 · INTRODUCTION

Why Habit Tracking?

40%

of daily actions are habits, not conscious decisions

66 days

average to form a new habit (Lally et al., UCL)

3×

higher success rate when habits are tracked daily

The Problem

- People struggle to maintain consistency in daily routines
- Existing tools are too complex, cluttered, or lack offline support
- No unified app that combines habit tracking + daily planning

Objectives

- Build a simple, beautiful habit tracking app
- Provide real-time progress & statistics
- Enable daily planning with a Planner module

Functional Requirements

Habit Management

Create, edit, delete habits with emoji, color, frequency, and reminder time

Daily Check-in

Mark habits as complete per day; view historical completion calendar

Reminders

Set daily notification reminders per habit with custom time picker

Statistics

Bar chart, heatmap, streak counters, and per-habit completion rate

Daily Planner

Add, edit, complete, and delete time-based tasks for any selected day

User Account

Register, login, logout with persisted profile name, bio, and settings

Non-Functional Requirements

Performance

- App launches in under 2 seconds
- SQLite queries execute < 50ms
- UI renders at 60fps on mid-range devices

Reliability

- All data persisted to SQLite — no data loss on restart
- Auth session survives app kill and relaunch
- Graceful error handling with inline messages

Usability

- Intuitive navigation — max 3 taps to any feature
- Clear visual hierarchy and readable typography
- Animated feedback on every user interaction

Security

- Credentials stored locally on device only
- Input validation on all form fields
- Session flag in SharedPreferences

03 · ARCHITECTURE

System Design & Architecture

MVVM + Repository Pattern

VIEW
Flutter Widgets / Screens
Home · Stats · Planner · Profile

↓

VIEW MODEL
ChangeNotifier Providers
HabitProvider · PlannerProvider · ProfileProvider

↓

REPOSITORY
AppDatabase (SQLite)
Habits · HabitLogs · PlannerTasks tables

↓

MODEL
Dart Data Classes
Habit · HabitLog · PlannerTask · UserProfile

Technology Stack

Flutter 3
Cross-platform UI framework

Dart
Programming language

SQLite (sqflite)
Local relational database

Provider
State management

SharedPreferences
Auth session & profile

Flutter Local Notif
Habit reminders

Database Design & ER Diagram

users	Account & profile
id (PK)	TEXT – UUID
name	TEXT NOT NULL
email	TEXT UNIQUE
password	TEXT NOT NULL
title	TEXT
bio	TEXT

habits	Habit definition
id (PK)	TEXT – UUID
name	TEXT NOT NULL
emoji	TEXT
color_value	INTEGER
freq	TEXT
reminder_time	TEXT (HH:mm)
streak	INTEGER DEFAULT 0
best_streak	INTEGER DEFAULT 0

habit_logs	Daily completion log
id (PK)	TEXT – UUID
habit_id	TEXT (FK → habits.id)
date	TEXT (ISO-8601)
completed	INTEGER 0/1
note	TEXT (optional)

planner_tasks	Daily scheduled tasks
id (PK)	TEXT – UUID
title	TEXT NOT NULL
subtitle	TEXT
time	TEXT (HH:mm)
color_value	INTEGER
date	TEXT (ISO-8601)
completed	INTEGER 0/1

ON DELETE CASCADE: deleting a habit automatically removes all its associated logs.

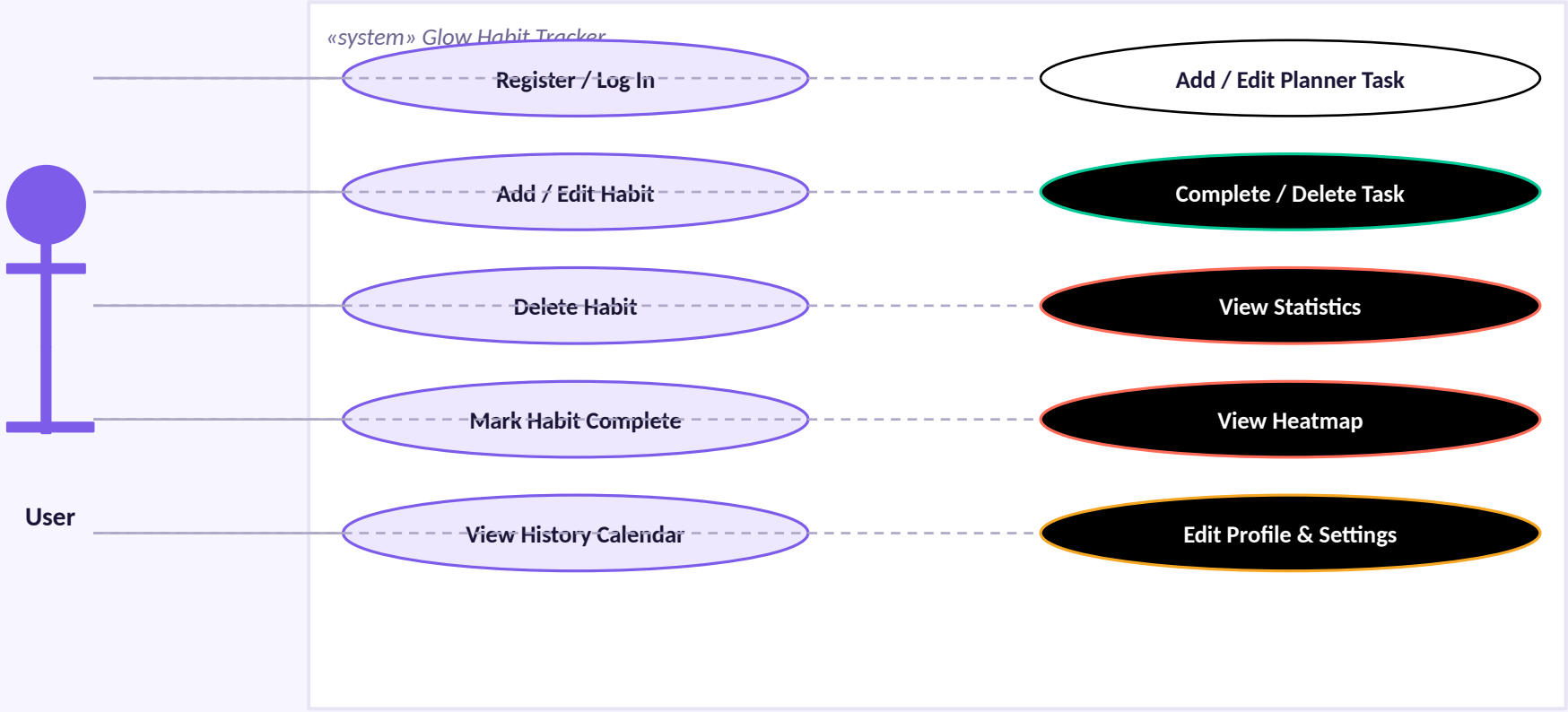
Relationships

users
1 — N → habits
One user owns many habits

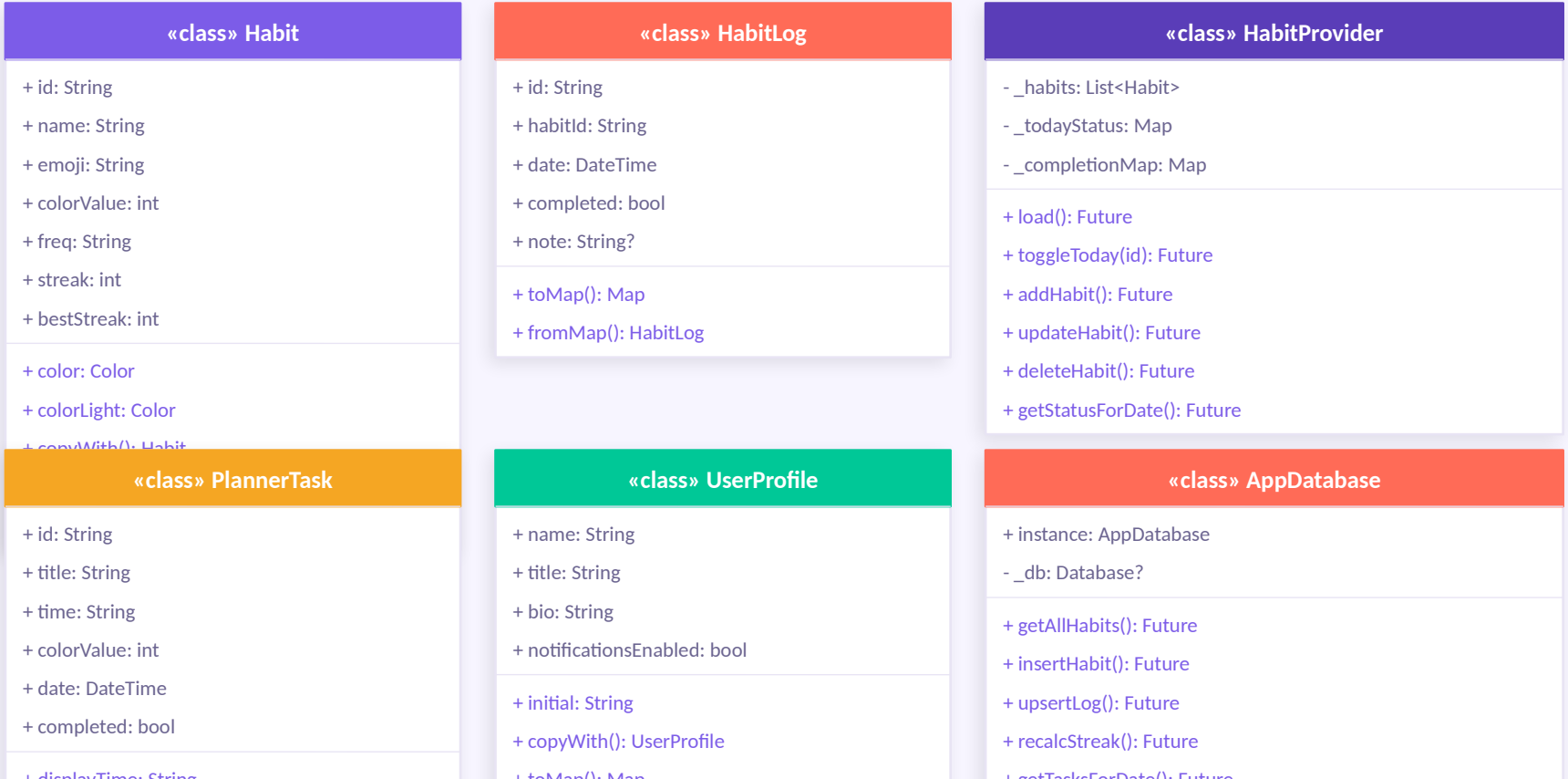
habits
1 — N → habit_logs
Each habit has many daily logs

users
1 — N → planner_tasks
One user has many planner tasks

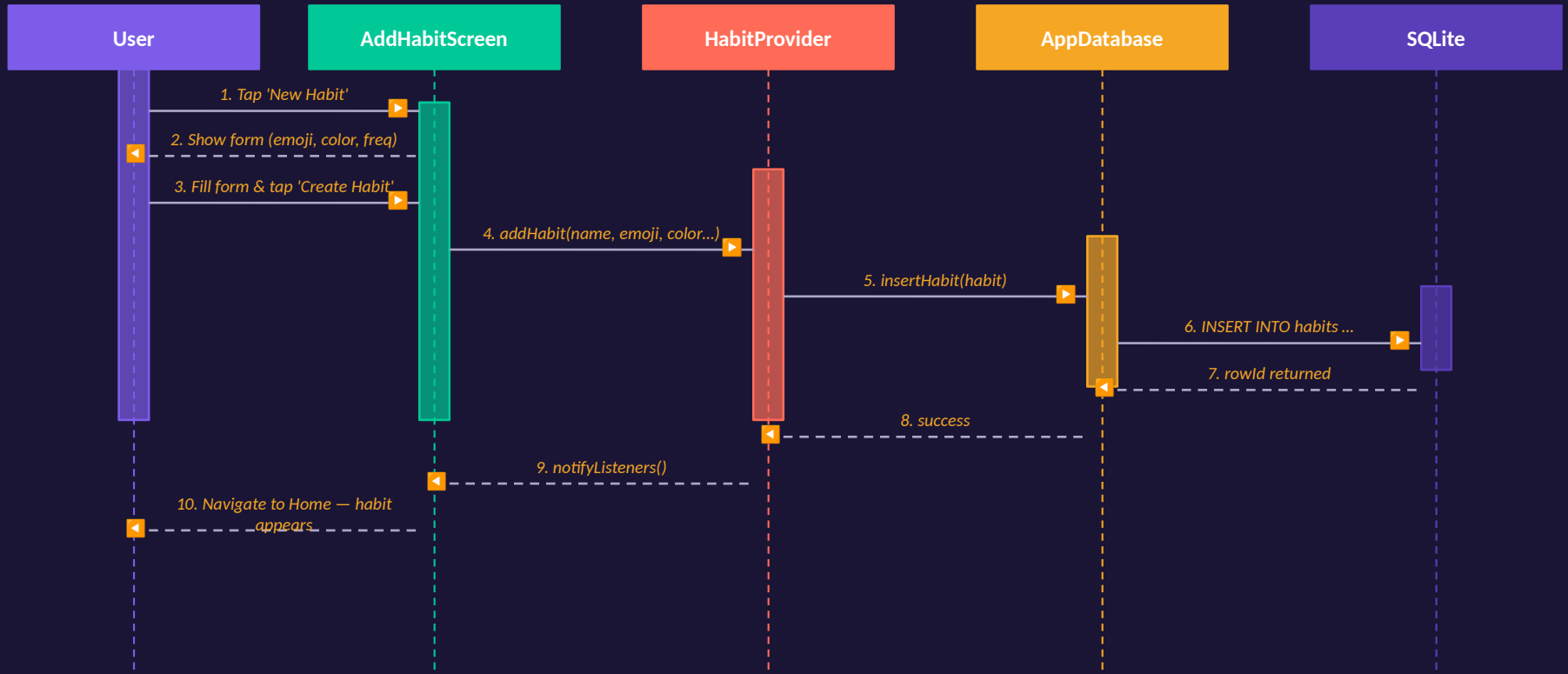
Use Case Diagram



Class Diagram



Sequence Diagram — Adding a Habit



Application Interface & UX

Home Screen

- 7-day week strip for date navigation
- Animated progress card with streak counter
- Habit tiles: emoji, freq chip, streak, toggle
- Toast notification on habit completion

Statistics

- 4 KPI cards: streak, completions, today, total
- 7-day bar chart from real DB data
- 13-week activity heatmap (91 cells)
- Per-habit progress bars with best streak

Daily Planner

- Horizontal 14-day scrollable date strip
- Add/edit tasks via bottom sheet
- Swipe left to delete tasks
- Complete toggle with strikethrough effect

Profile & Auth

- Splash → Sign Up or Log In screens
- Full input validation & error banners
- Edit name, title, bio (persisted)
- Notifications & Weekly Report toggles

UX Design Principles: light mode palette · animated transitions · consistent card shadows · bottom-sheet forms · empty-state illustrations

07 · DEMO

User Demo Scenario

Step-by-step: A new user starts and tracks their first habit

01 — Launch App

Splash screen appears — user taps Get Started

02 — Create Account

Fills name, email, password — taps Create Account

03 — Home Screen

Sees welcome greeting with their name in 'Hey, Alex!'

04 — Add First Habit

Taps + button → selects emoji 🏃, purple color, Daily freq

05 — Check Habit

Taps the circle → habit turns green, toast pops up

06 — Check Stats

Navigates to Stats tab — sees 1/1 done, streak = 1

07 — Plan the Day

Opens Planner → adds '7:30 AM Meditation' task

08 — Edit Profile

Goes to Profile → edits bio, toggles notifications on

Challenges & Solutions



Streak Calculation Accuracy

Problem: Streak logic broke when users skipped days or toggled in the past

Solution: Rebuilt using recalcStreak() — queries full log history, walks back from today/yesterday counting consecutive completed dates



Reactive Name in Header

Problem: Home screen showed hardcoded 'Alex' — not dynamic

Solution: Used context.watch<ProfileProvider>() inside Consumer and passed name to _Header widget — rebuilds instantly on profile edit



State Sync After DB Writes

Problem: UI not updating after DB insert because provider wasn't refreshing

Solution: Added notifyListeners() after every DB operation + refreshed completionMap and todayStatus together atomically



Auth Flow & Navigation

Problem: Logging out left the navigator stack in a broken state

Solution: Introduced AuthGate widget that watches isLoggedIn — automatically swaps between SplashScreen and Shell without manual navigation

08 · CONCLUSION

Conclusion & Future Work

What We Achieved

- Full MVVM architecture with Provider state management
- SQLite database with 3 tables and ON DELETE CASCADE
- Real-time streak calculation from log history
- Complete auth flow: sign up, login, logout, session persistence
- Dynamic profile that reflects across all screens instantly
- 13-week activity heatmap driven by real DB data

Future Improvements

- 🔔 Real push notifications via flutter_local_notifications
- ☁️ Cloud sync with Firebase (multi-device support)
- 🌙 Full dark mode implementation
- 🤝 Social features: share streaks, challenges with friends
- 🤖 AI habit suggestions based on user behavior patterns
- 📱 iOS build + App Store publication



Thank You

Questions & Discussion

Flutter | SQLite | Provider | MVVM | Dart

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